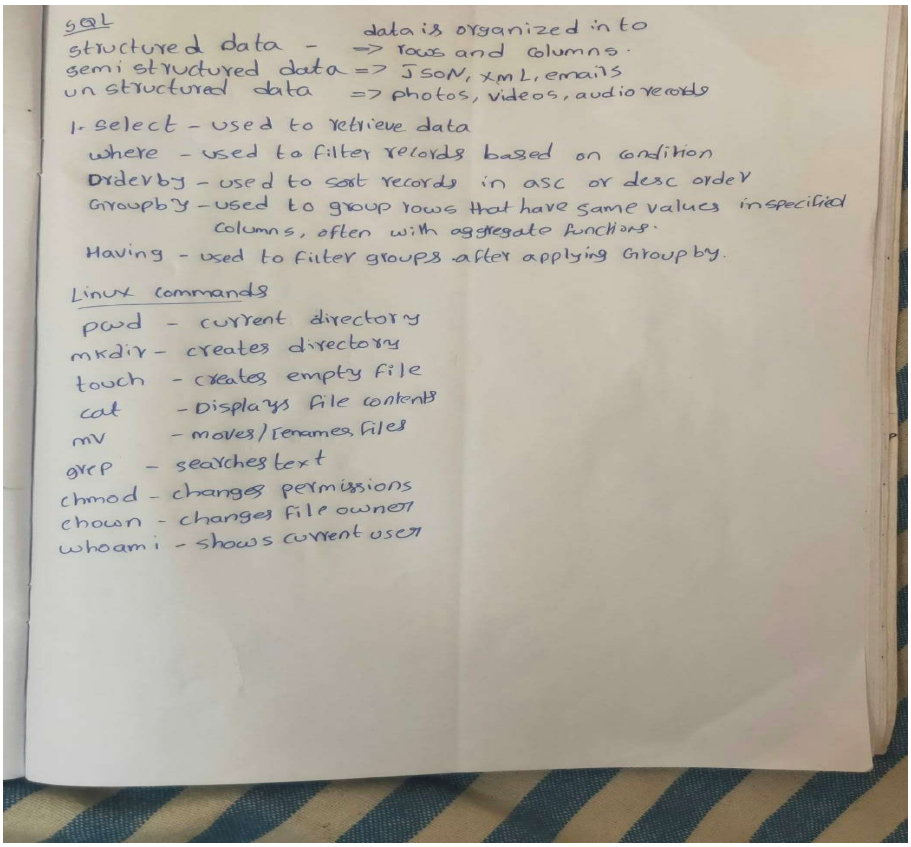
JAIN UNIVERSITY

DEEMED-TO-BE UNIVERSITY

AWS CLOUD & DATA ENGINEERING



Name	:	Vempalli Jabivulla
Roll Number	:	23BTRCL237
College Name	:	Jain University
Batch	:	2027
Date	:	11-10-2026



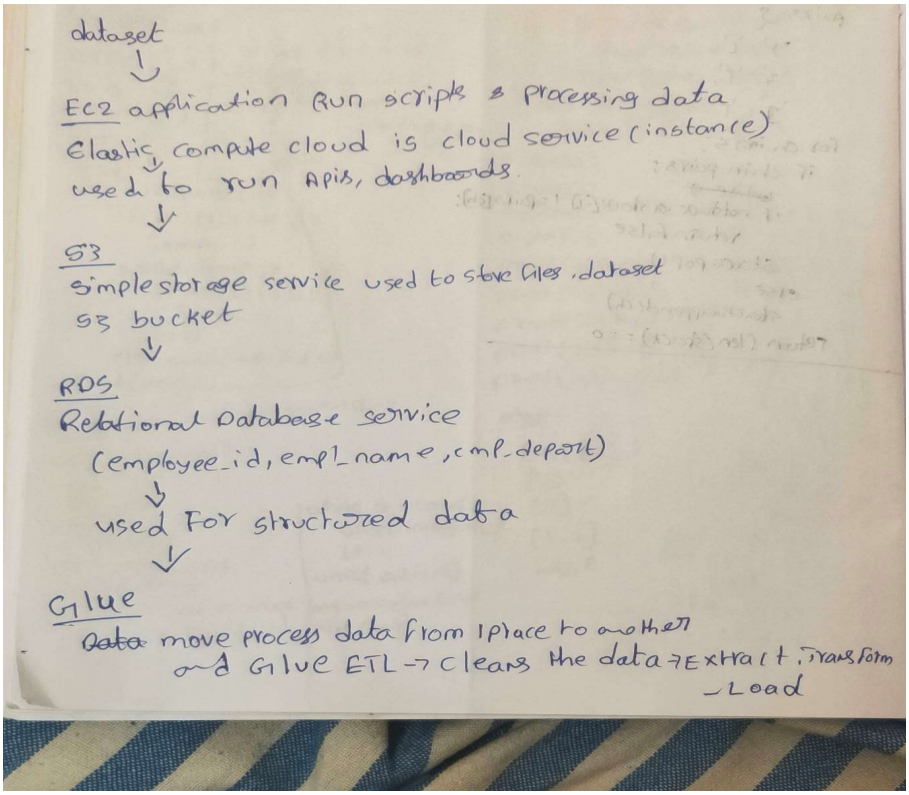
SQL

structured data - data is organized in to => rows and columns.
semi structured data => JSON, XML, emails
un structured data => photos, videos, audio records

1. select - used to retrieve data
   where - used to filter records based on condition
   Orderby - used to sort records in asc or desc order
   Groupby - used to group rows that have same values in specified columns, often with aggregate functions.
   Having - used to filter groups after applying Groupby.

Linux commands

pwd - current directory
mkdir - creates directory
touch - creates empty file
cat - Displays file contents
mv - moves/renames files
grep - searches text
chmod - changes permissions
chown - changes file owner
whoami - shows current user



dataset
↓
EC2 application Run scripts & processing data
Elastic compute cloud is cloud service (instance)
used to run APIs, dashboards.
↓
S3
simple storage service used to store files, dataset
s3 bucket
↓
RDS
Relational Database service
(employee_id, emp1_name, emp_depart)
↓
used For structured data
↓
Glue
Data move process data from 1place to another
→ A Glue ETL → Cleans the data → Extract, Transform - Load

SQL COMMANDS

1.

```
25
26  • SELECT * FROM employee;
27
28
29
30
```

Output | Result #5

employee_id	name	department	salary	city
1	Ravi	IT	60000	Bangalore
2	Priya	HR	35000	Chennai
3	Amit	IT	75000	Bangalore
4	Sneha	Finance	45000	Mumbai
5	Rahul	HR	40000	Chennai
6	Kiran	IT	65000	Hyderabad
7	Anjali	Finance	55000	Bangalore
8	Vijay	Marketing	38000	Mumbai
9	Pooja	IT	80000	Chennai
10	Arun	Marketing	42000	Hyderabad

2.) count employees in each department

```
27
28  • SELECT department,COUNT(*) AS employee_count
29    FROM employee
30    GROUP BY department;
31
32
33
```

Output | Result #6

department	employee_count
IT	4
HR	2
Finance	2
Marketing	2

3.) find department having more than 2 employees

```
32
33  SELECT department, COUNT(*) AS employee_count
34  FROM employee
35  GROUP BY department
36  HAVING count(*)>2;
37
38
```

Output | Result #7

department	employee_count
IT	4

4.)find 2nd highest salary

```
38
39  SELECT MAX(salary) AS second_highest_salary
40  FROM employee
41  WHERE salary < (SELECT MAX(salary) FROM employee);
42
43
```

Output | Result #8

second_highest_salary
75000

5.)employee earning more than avg salary

```
43  SELECT *
44  FROM employee
45  WHERE salary > (SELECT AVG(salary)FROM employee);
46
47
```

Output | Result #9

employee_id	name	department	salary	city
1	Ravi	IT	60000	Bangalore
3	Amit	IT	75000	Bangalore
6	Kiran	IT	65000	Hyderabad
7	Anjali	Finance	55000	Bangalore
9	Pooja	IT	80000	Chennai

6.) ORDERBY

```
47
48  SELECT * FROM employee
49  ORDER BY salary DESC;
50
```

Output | Result #10

employee_id	name	department	salary	city
9	Pooja	IT	80000	Chennai
3	Amit	IT	75000	Bangalore
6	Kiran	IT	65000	Hyderabad
1	Ravi	IT	60000	Bangalore
7	Anjali	Finance	55000	Bangalore
4	Sneha	Finance	45000	Mumbai
10	Arun	Marketing	42000	Hyderabad
5	Rahul	HR	40000	Chennai
8	Vijay	Marketing	38000	Mumbai
2	Priya	HR	35000	Chennai

7.) Join

```
70  SELECT
71      e.name,
72      e.department,
73      d.city
74  FROM employee e
75  JOIN departments d
76  ON e.department = d.department;
```

Output | Result #13

name	department	city
Pooja	IT	Bangalore
Kiran	IT	Bangalore
Amit	IT	Bangalore
Ravi	IT	Bangalore
Rahul	HR	Chennai
Priya	HR	Chennai
Anjali	Finance	Mumbai
Sneha	Finance	Mumbai
Arun	Marketing	Pune
Vijay	Marketing	Pune

8.) Left join

```
78  • SELECT e.name, e.department, d.city
79    FROM employee e
80    LEFT JOIN departments d
81    ON e.department = d.department;
```

Output | Result #14

name	department	city
Ravi	IT	Bangalore
Priya	HR	Chennai
Amit	IT	Bangalore
Sneha	Finance	Mumbai
Rahul	HR	Chennai
Kiran	IT	Bangalore
Anjali	Finance	Mumbai
Vijay	Marketing	Pune
Pooja	IT	Bangalore
Arun	Marketing	Pune

9.) Right Join

```
83  • SELECT e.name, e.department, d.city
84    FROM employee e
85    RIGHT JOIN departments d
86    ON e.department = d.department;
```

Output | Result #15

name	department	city
Pooja	IT	Bangalore
Kiran	IT	Bangalore
Amit	IT	Bangalore
Ravi	IT	Bangalore
Rahul	HR	Chennai
Priya	HR	Chennai
Anjali	Finance	Mumbai
Sneha	Finance	Mumbai
NULL	NULL	Hyderabad
Arun	Marketing	Pune

10.) Union

```sql
SELECT e.name, e.department, d.city
FROM employee e
LEFT JOIN departments d
ON e.department = d.department

UNION

SELECT e.name, d.department, d.city
FROM employee e
RIGHT JOIN departments d
ON e.department = d.department;
```

Output | Result #16

name	department	city
Ravi	IT	Bangalore
Priya	HR	Chennai
Amit	IT	Bangalore
Sneha	Finance	Mumbai
Rahul	HR	Chennai
Kiran	IT	Bangalore
Anjali	Finance	Mumbai
Vijay	Marketing	Pune
Pooja	IT	Bangalore
Arun	Marketing	Pune

11.) Maximum salary of employees in the HR department.

```sql
SELECT *
FROM employee
WHERE salary > (
    SELECT MAX(salary)
    FROM employee
    WHERE department = 'HR'
);9
```

Output | Result #17

employee_id	name	department	salary	city
1	Ravi	IT	60000	Bangalore
3	Amit	IT	75000	Bangalore
4	Sneha	Finance	45000	Mumbai
6	Kiran	IT	65000	Hyderabad
7	Anjali	Finance	55000	Bangalore
9	Pooja	IT	80000	Chennai
10	Arun	Marketing	42000	Hyderabad

12.) Group by

```
108  SELECT department, COUNT(*) AS total_employees
109  FROM employee
110  GROUP BY department;
```

Output | Result #18

department	total_employees
IT	4
HR	2
Finance	2
Marketing	2

13.) total salary of each department using GROUP BY.

```
111
112  SELECT department, SUM(salary) AS total_salary
113  FROM employee
114  GROUP BY department;
```

Output | Result #19

department	total_salary
IT	280000
HR	75000
Finance	100000
Marketing	80000

14.) Employees earning more than 40,000, sorted by salary in descending order.

```
115
116  SELECT *
117  FROM employee
118  WHERE salary > 40000
119  ORDER BY salary DESC;
```

Output | Result #20

employee_id	name	department	salary	city
9	Pooja	IT	80000	Chennai
3	Amit	IT	75000	Bangalore
6	Kiran	IT	65000	Hyderabad
1	Ravi	IT	60000	Bangalore
7	Anjali	Finance	55000	Bangalore
4	Sneha	Finance	45000	Mumbai
10	Arun	Marketing	42000	Hyderabad

15.) Employees from the IT and HR departments, sorted alphabetically by name

```
121  • SELECT *
122    FROM employee
123    WHERE department IN ('IT', 'HR')
124    ORDER BY name ASC;
```

Output | Result #21

employee_id	name	department	salary	city
3	Amit	IT	75000	Bangalore
6	Kiran	IT	65000	Hyderabad
9	Pooja	IT	80000	Chennai
2	Priya	HR	35000	Chennai
5	Rahul	HR	40000	Chennai
1	Ravi	IT	60000	Bangalore

LINUX COMMANDS

Basic commands

```
[ec2-user@ip-172-31-21-235 ~]$ PWD
-bash: PWD: command not found
[ec2-user@ip-172-31-21-235 ~]$
```

```
[ec2-user@ip-172-31-21-235 ~]$ ls
index.html.save
```

```
[ec2-user@ip-172-31-21-235 ~]$ ls -l
total 4
-rw-------. 1 root root 1548 Oct 10 05:06 index.html.save
```

```
[ec2-user@ip-172-31-21-235 ~]$ ls -la
total 20
drwx------. 3 ec2-user ec2-user  118 Oct 10 05:06 .
drwxr-xr-x. 3 root     root       22 Oct 10 04:53 ..
-rw-------. 1 ec2-user ec2-user  806 Oct 11 09:29 .bash_history
-rw-r--r--. 1 ec2-user ec2-user   18 Jan 28  2023 .bash_logout
-rw-r--r--. 1 ec2-user ec2-user  141 Jan 28  2023 .bash_profile
-rw-r--r--. 1 ec2-user ec2-user  492 Jan 28  2023 .bashrc
drwx------. 2 ec2-user ec2-user   29 Oct 10 04:53 .ssh
-rw-------. 1 root     root     1548 Oct 10 05:06 index.html.save
[ec2-user@ip-172-31-21-235 ~]$
```

File and directory management

```
[ec2-user@ip-172-31-21-235 ~]$ mkdir -p linux_practice
[ec2-user@ip-172-31-21-235 ~]$ cd linux_practice
[ec2-user@ip-172-31-21-235 linux_practice]$ touch file1.txt
[ec2-user@ip-172-31-21-235 linux_practice]$ echo "Linux Practical" > file1.txt
[ec2-user@ip-172-31-21-235 linux_practice]$ cat file1.txt
Linux Practical
[ec2-user@ip-172-31-21-235 linux_practice]$
```

File operations

```
[ec2-user@ip-172-31-21-235 linux_practice]$ cp file1.txt file2.txt
[ec2-user@ip-172-31-21-235 linux_practice]$ mv file2.txt renamed.txt
[ec2-user@ip-172-31-21-235 linux_practice]$ ls -l
total 8
-rw-r--r--. 1 ec2-user ec2-user 16 Oct 11 09:33 file1.txt
-rw-r--r--. 1 ec2-user ec2-user 16 Oct 11 09:35 renamed.txt
[ec2-user@ip-172-31-21-235 linux_practice]$ head -n 5 file1.txt
Linux Practical
[ec2-user@ip-172-31-21-235 linux_practice]$ tail -n 5 file1.txt
Linux Practical
[ec2-user@ip-172-31-21-235 linux_practice]$
```

Text-processing commands

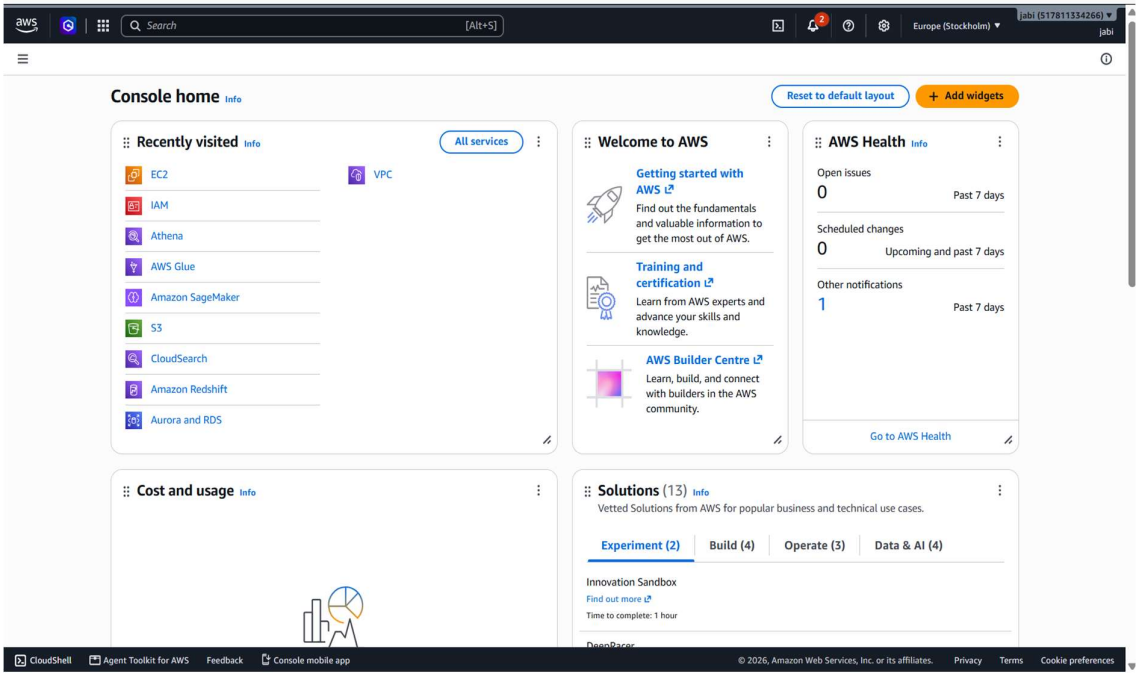
```
[ec2-user@ip-172-31-21-235 linux_practice]$ grep "Linux" file1.txt
Linux Practical
[ec2-user@ip-172-31-21-235 linux_practice]$ wc -l file1.txt
1 file1.txt
[ec2-user@ip-172-31-21-235 linux_practice]$ echo "Commands Practice" >> file1.txt
[ec2-user@ip-172-31-21-235 linux_practice]$ cat file1.txt
Linux Practical
Commands Practice
[ec2-user@ip-172-31-21-235 linux_practice]$
```

nginx

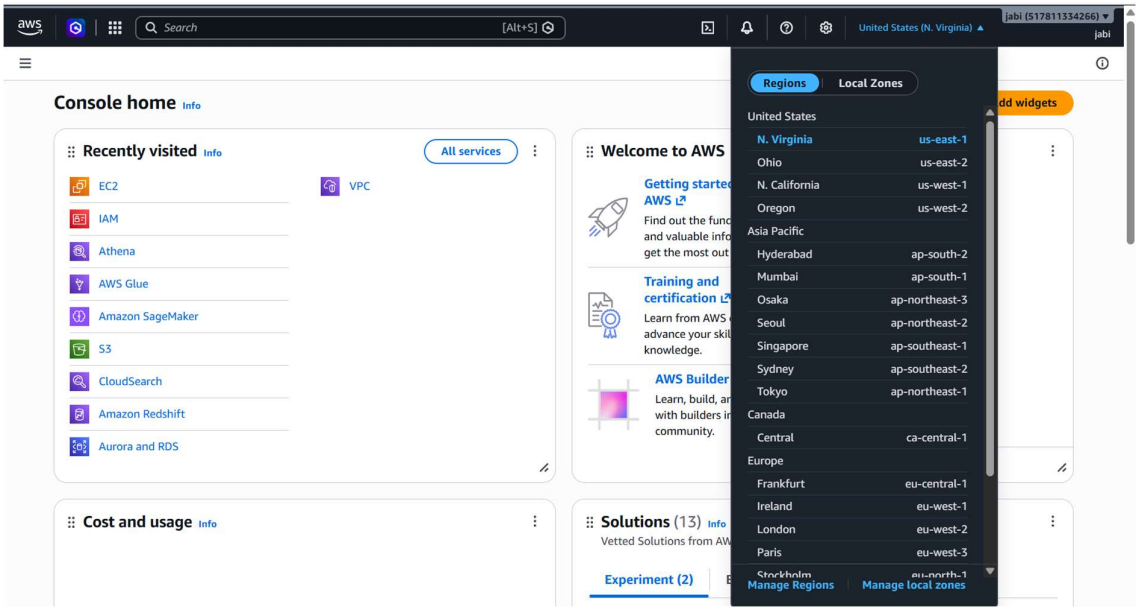
```
[ec2-user@ip-172-31-21-235 linux_practice]$ sudo systemctl start nginx
[ec2-user@ip-172-31-21-235 linux_practice]$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
[ec2-user@ip-172-31-21-235 linux_practice]$ nginx -v
nginx version: nginx/1.30.5
[ec2-user@ip-172-31-21-235 linux_practice]$
```

AWS ACCOUNT/REGION

AWS Console

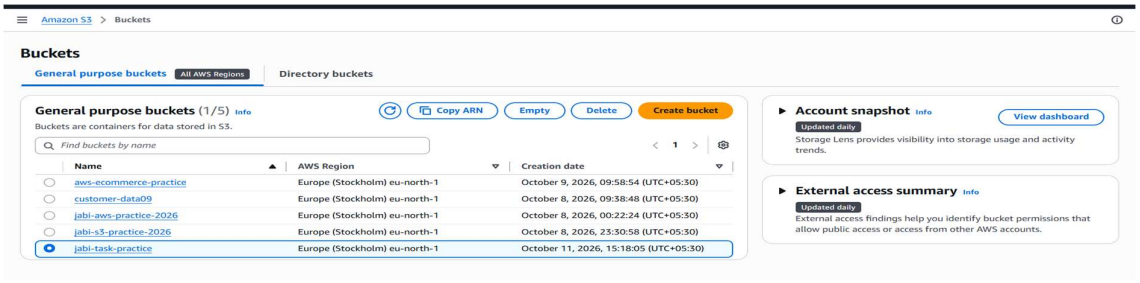


Selected AWS Region

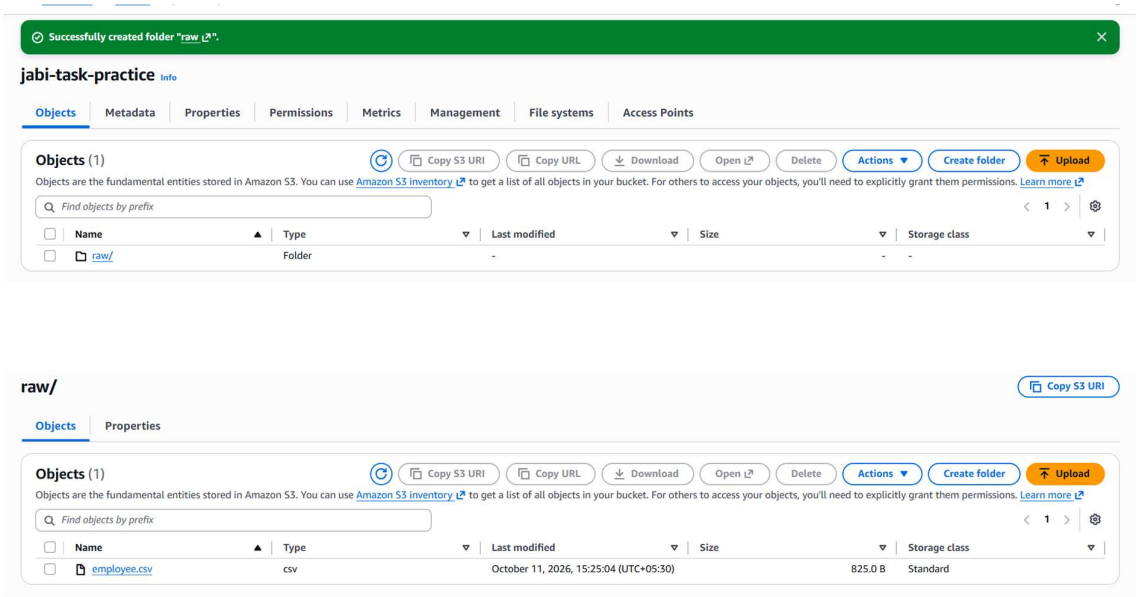


AMAZON S3

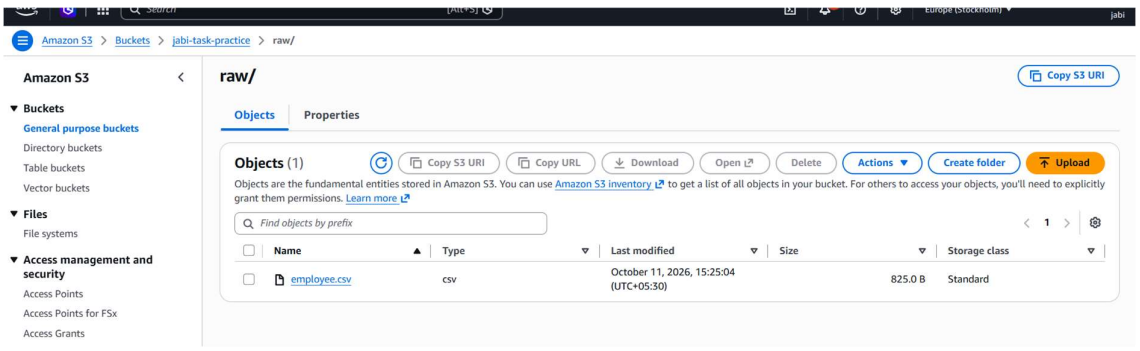
S3 bucket created



raw/ folder

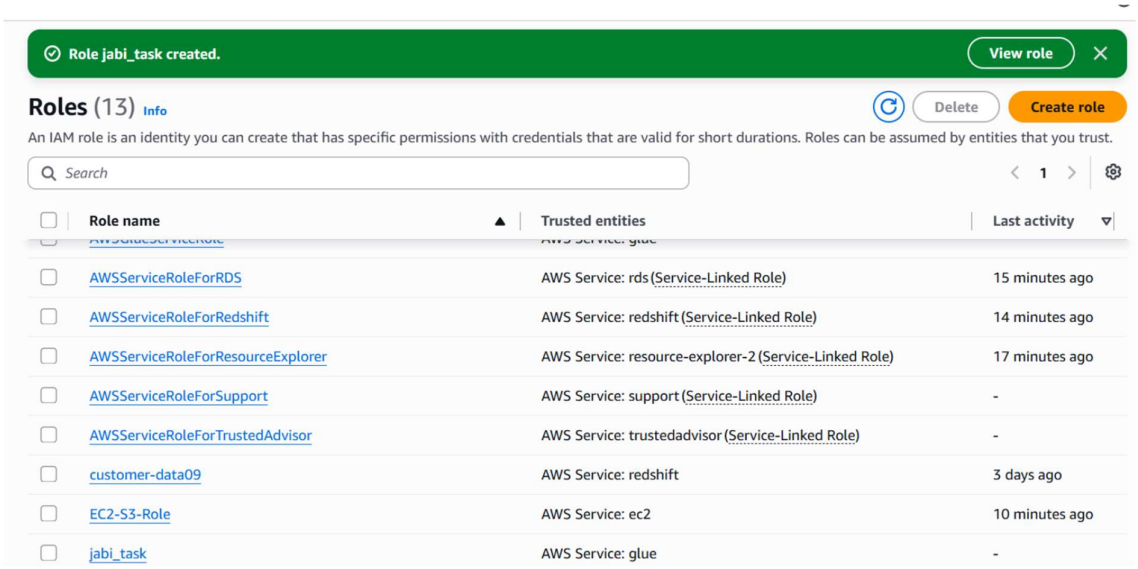


employees.csv uploaded inside raw

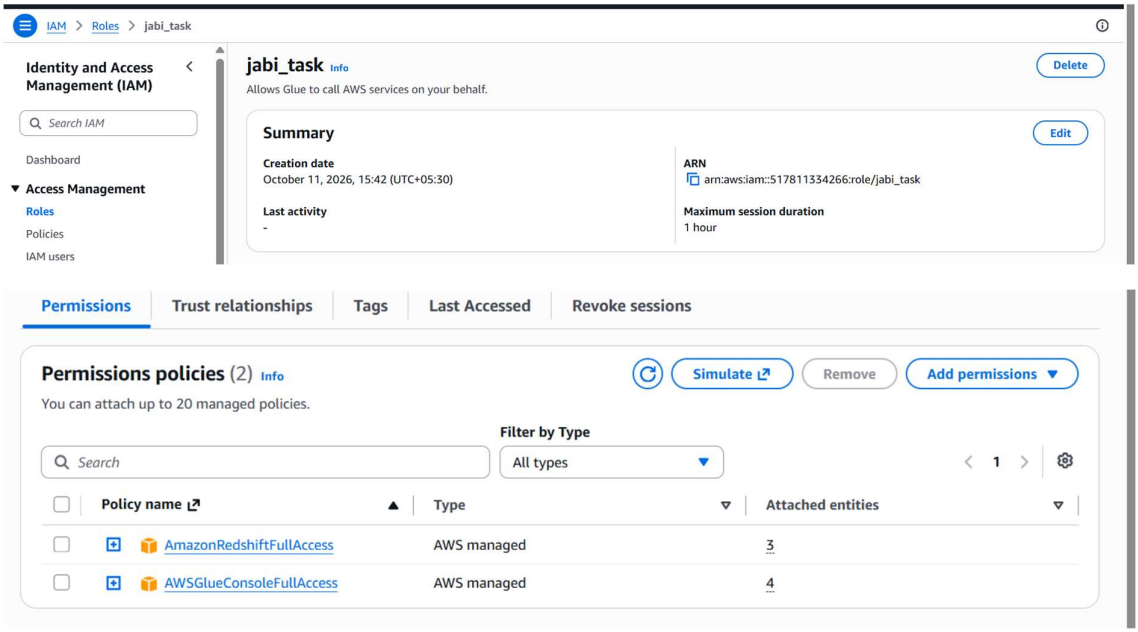


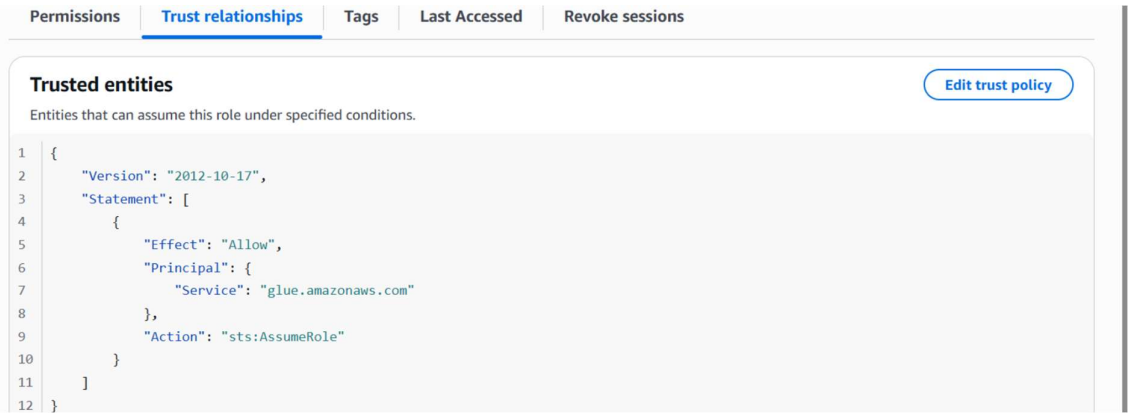
IAM

IAM Role used for AWS Glue



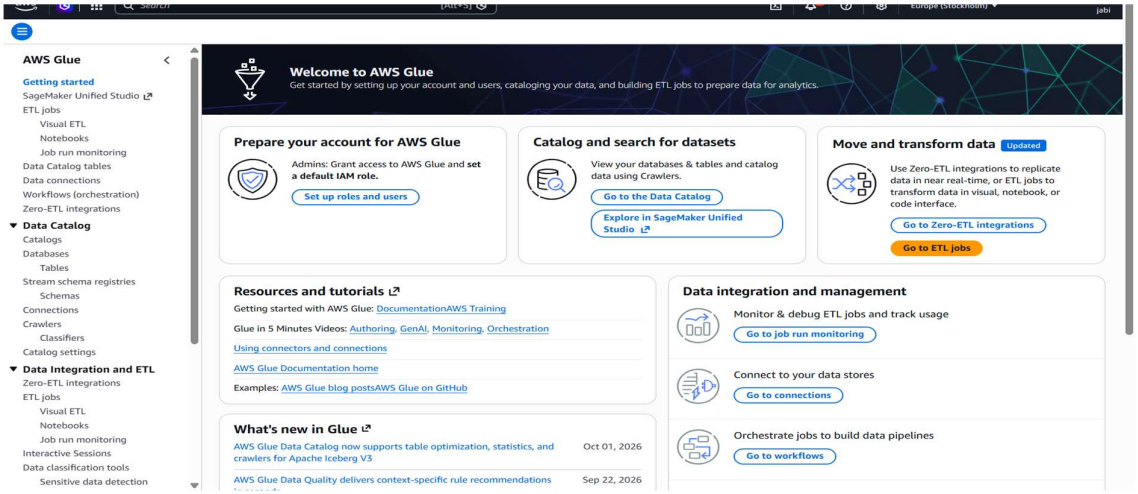
Relevant permissions/policies



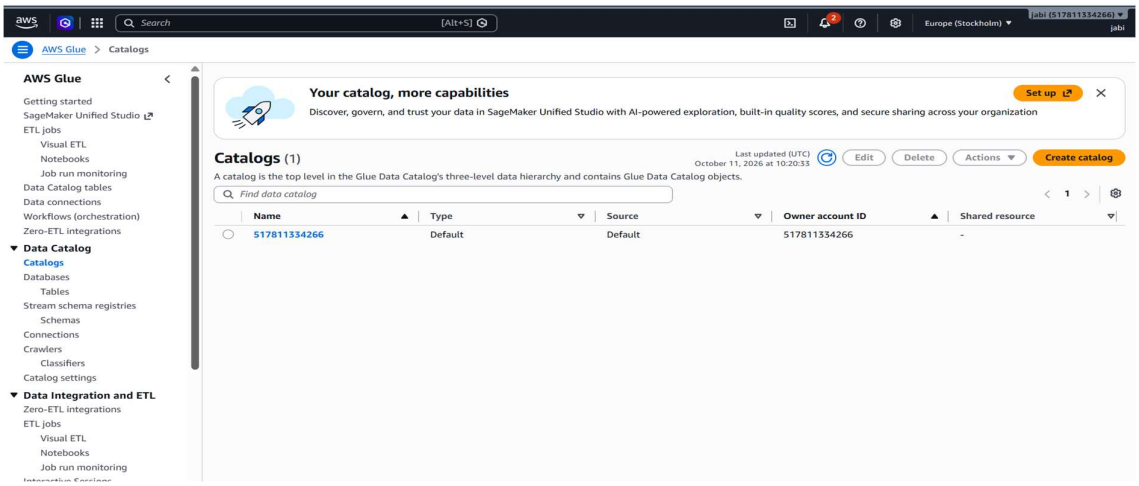


AWS GLUE DATABASE

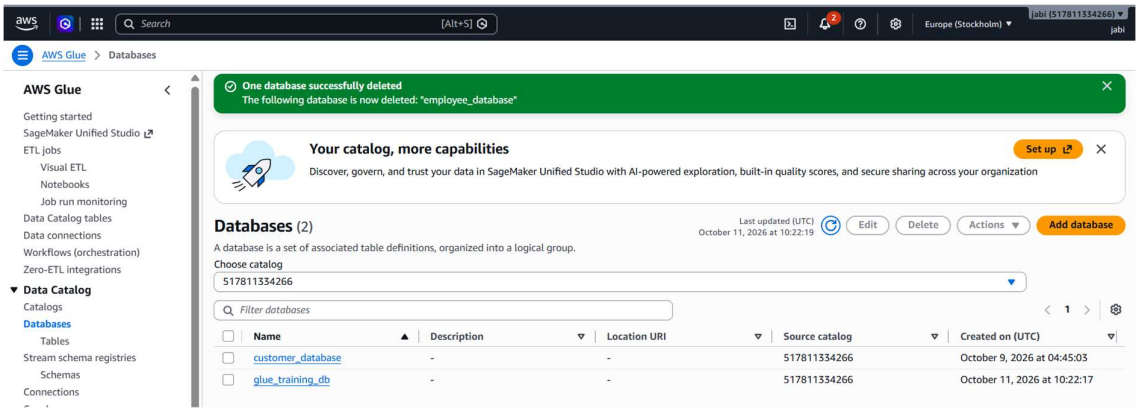
AWS Glue Console



Data Catalog

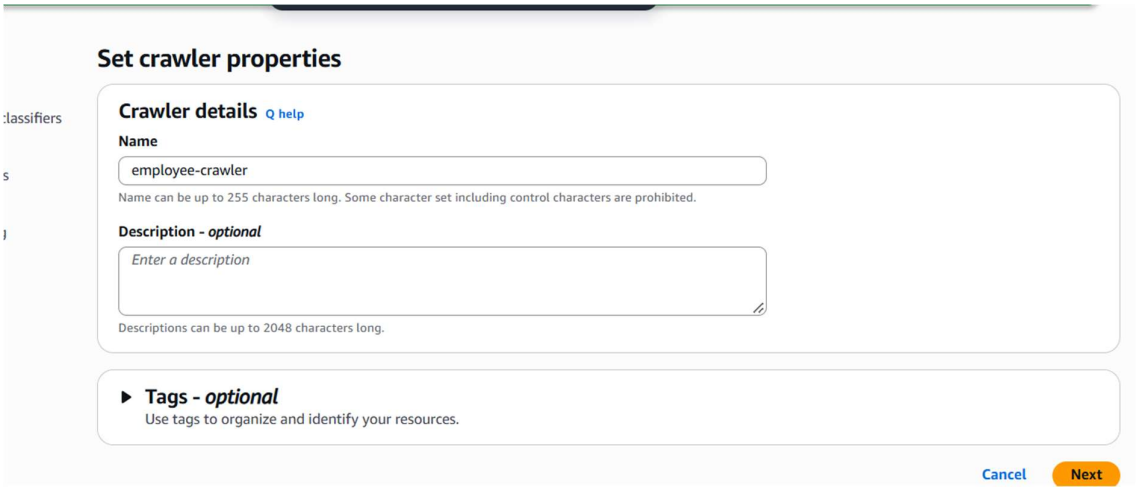


Database created

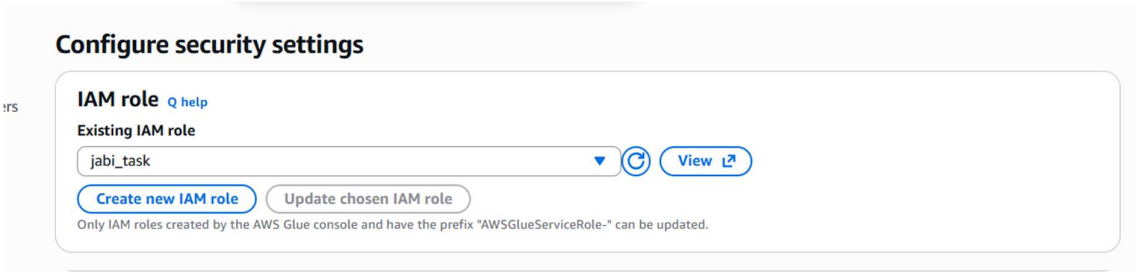


GLUE CRAWLER

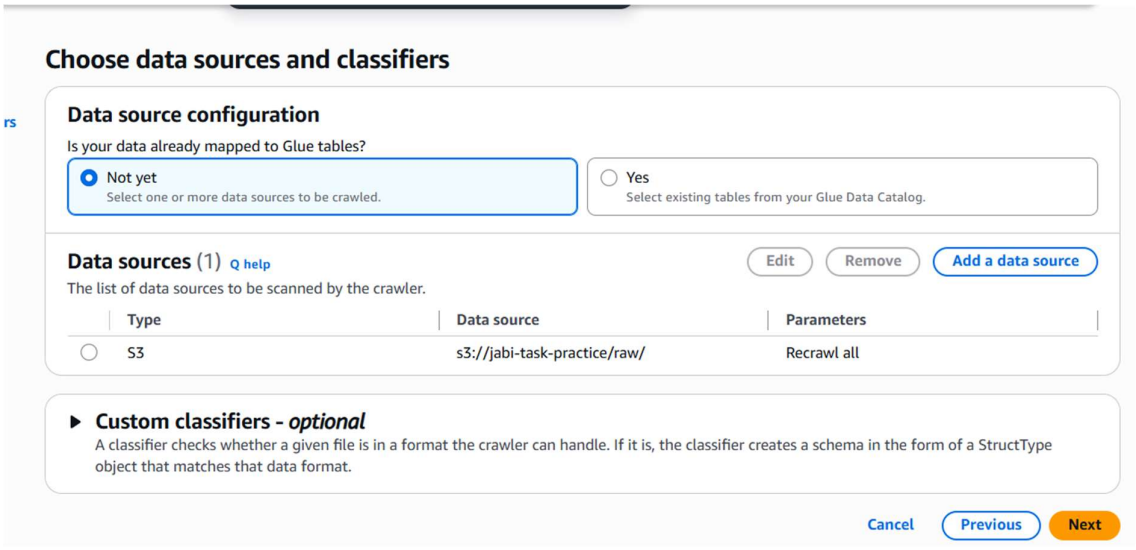
Crawler name



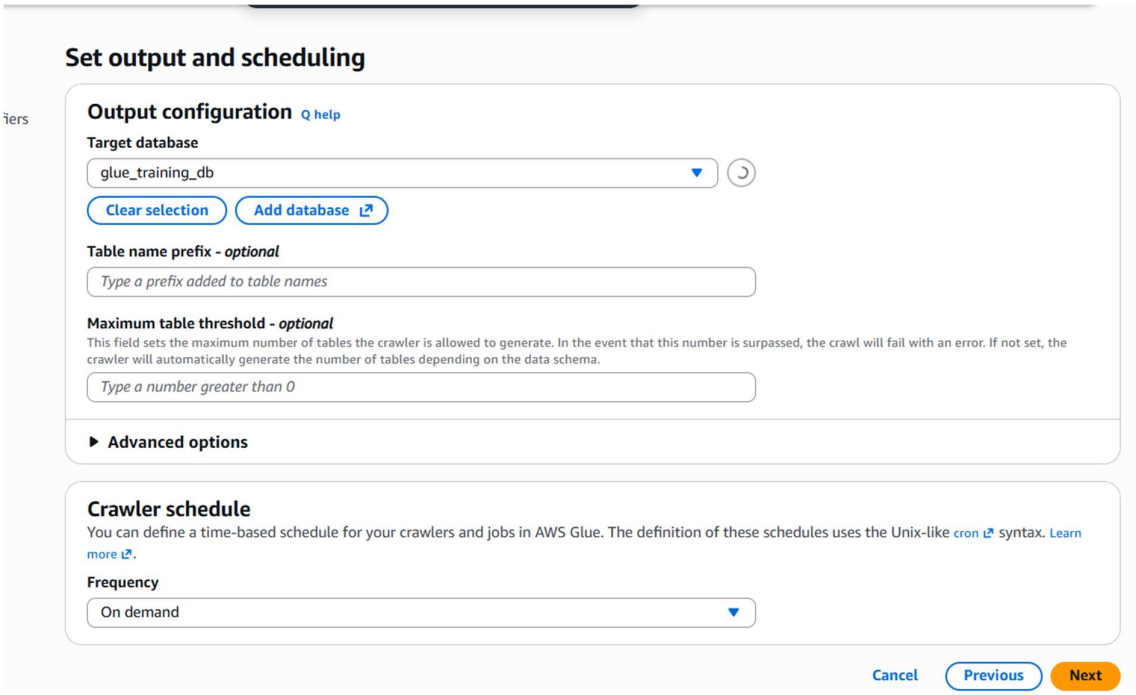
IAM role



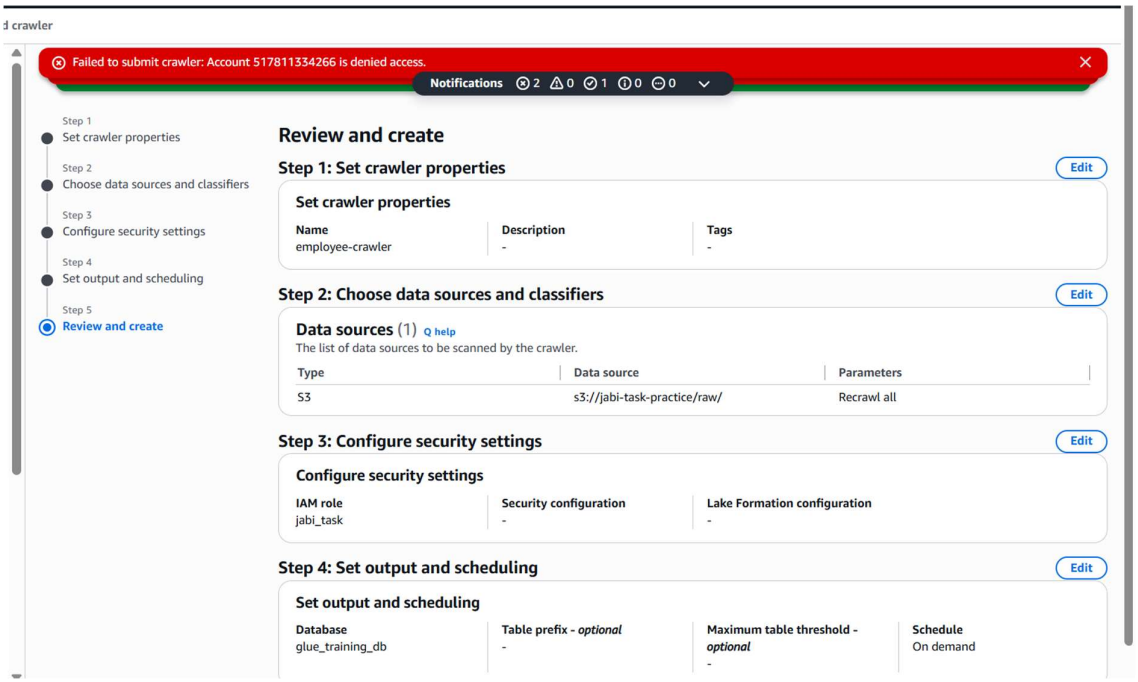
S3 source location



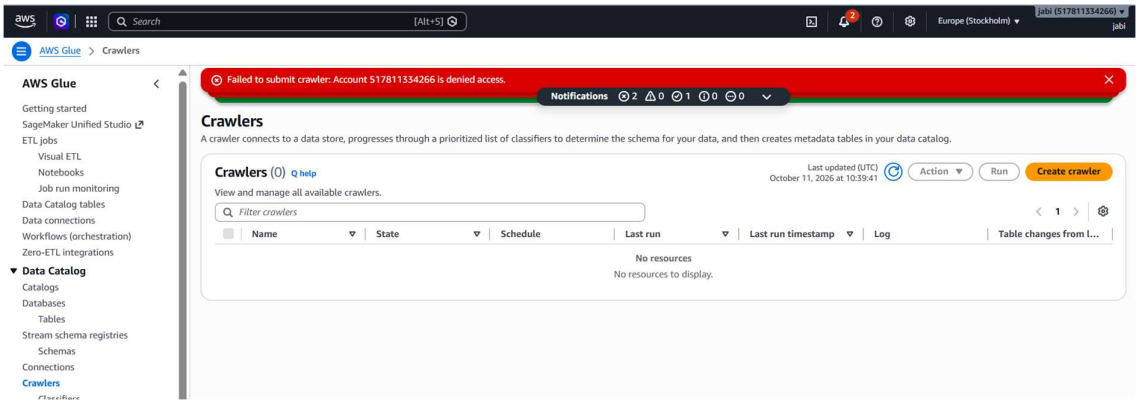
Target database



Crawler configuration



Crawler Execution



Glue Data Catalog

Database

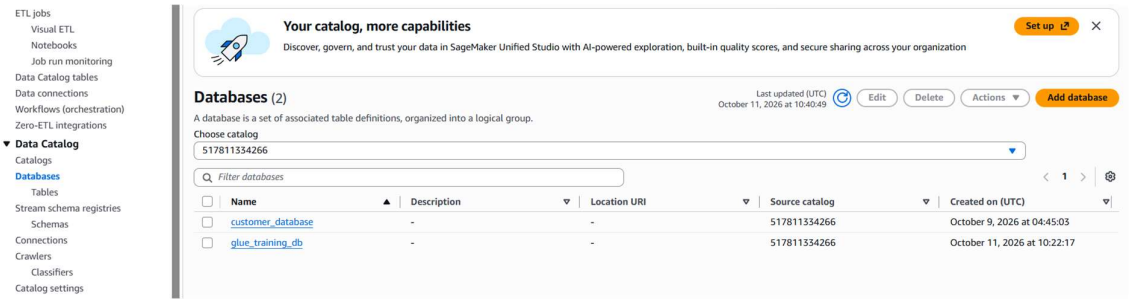


Table successfully created

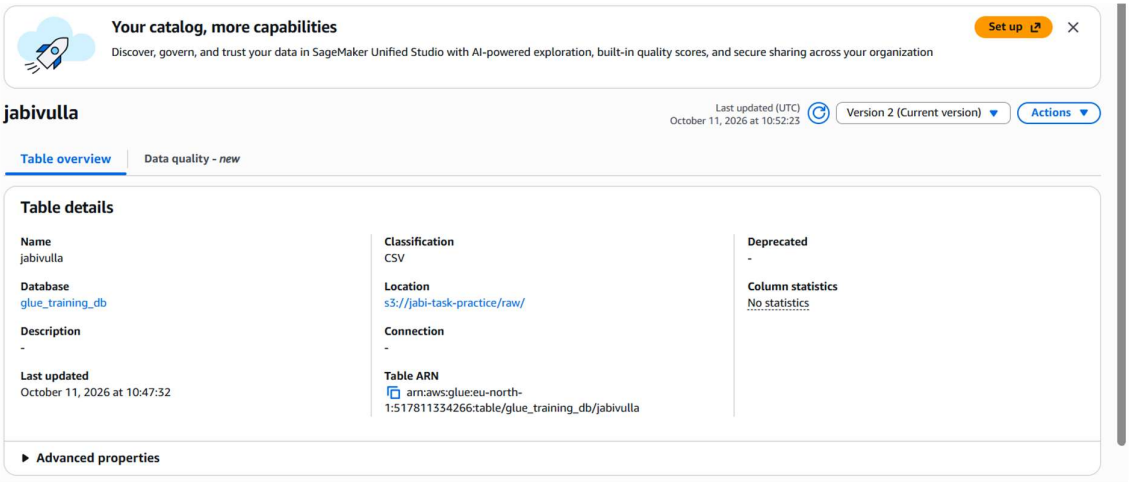
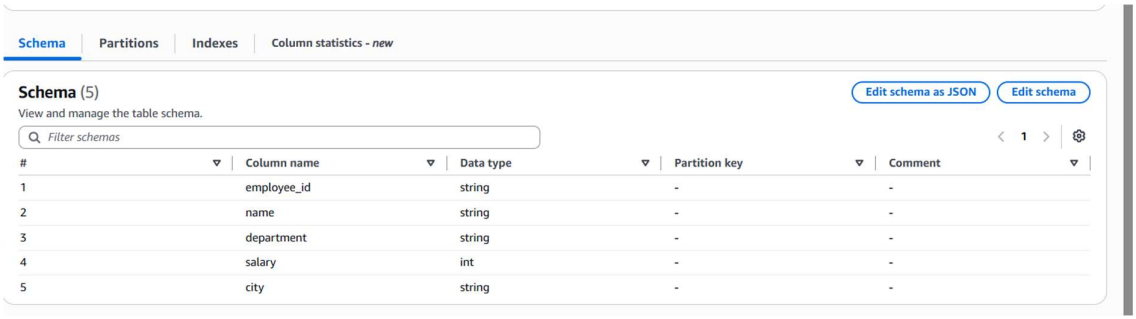


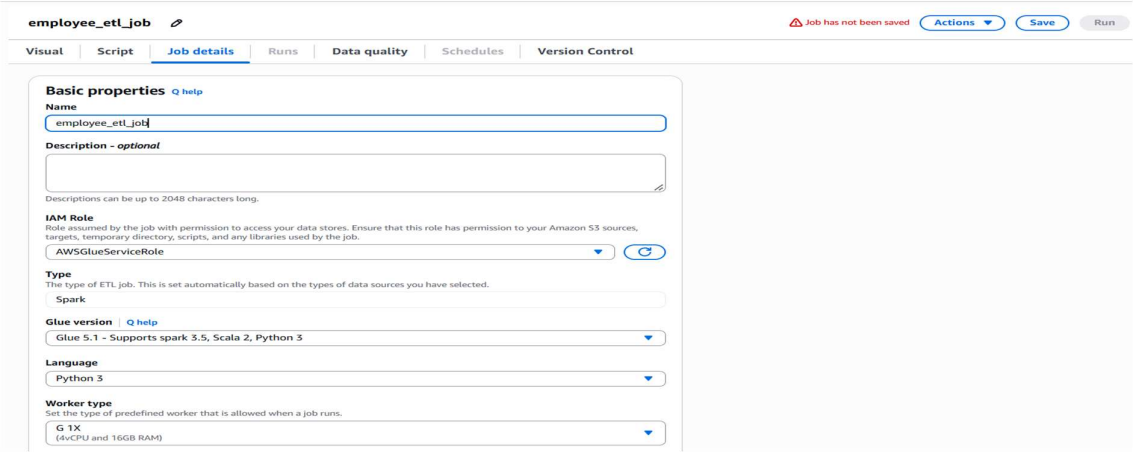
TABLE SCHEMA



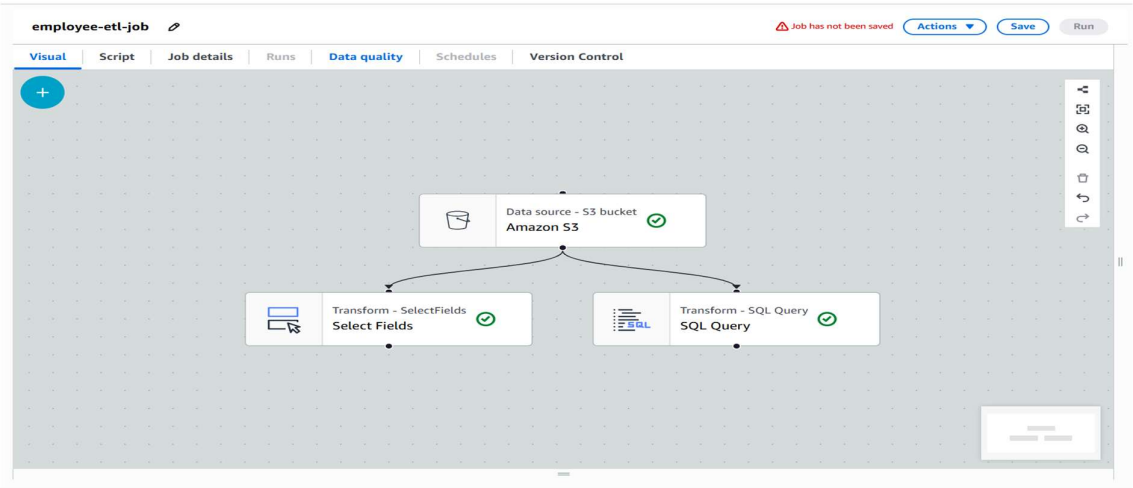
GLUE VISUAL ETL

Show the complete workflow:

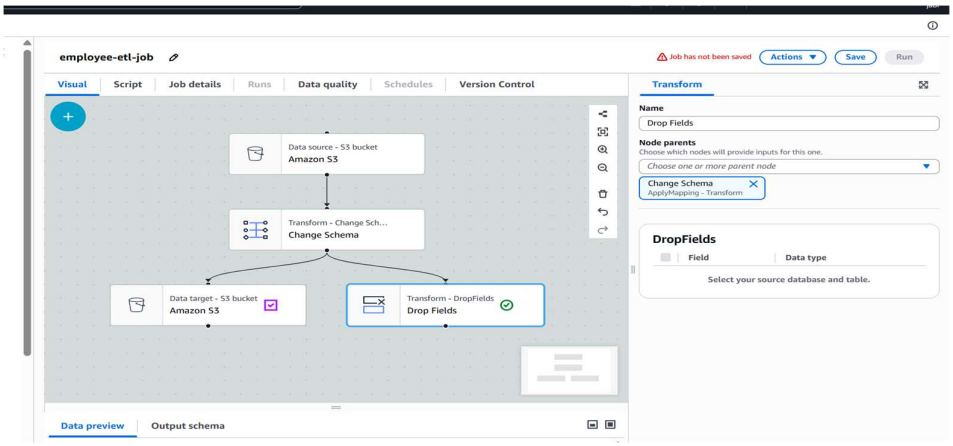
The entire Visual ETL canvas should be visible.



Data Catalog → employees → Transformation → S3 Target

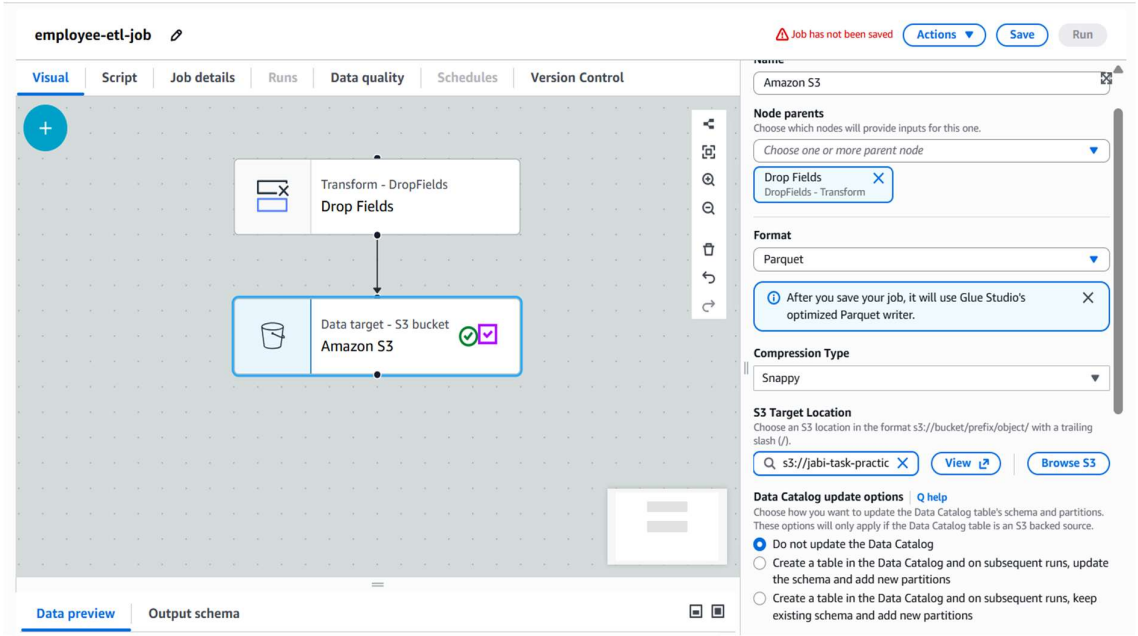


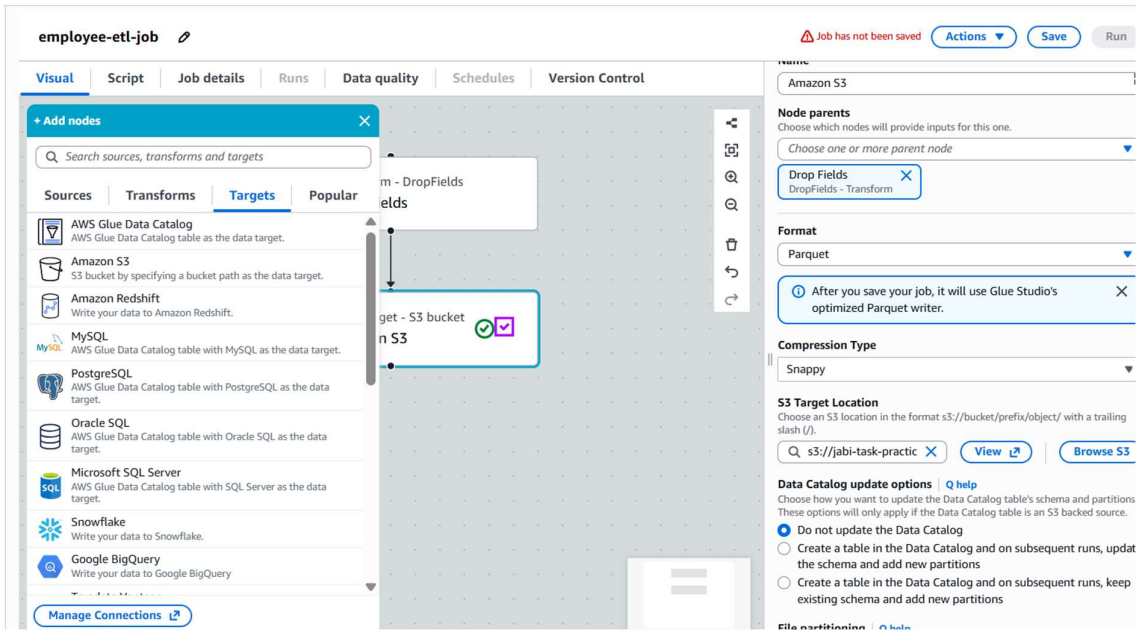
Glue Transformation



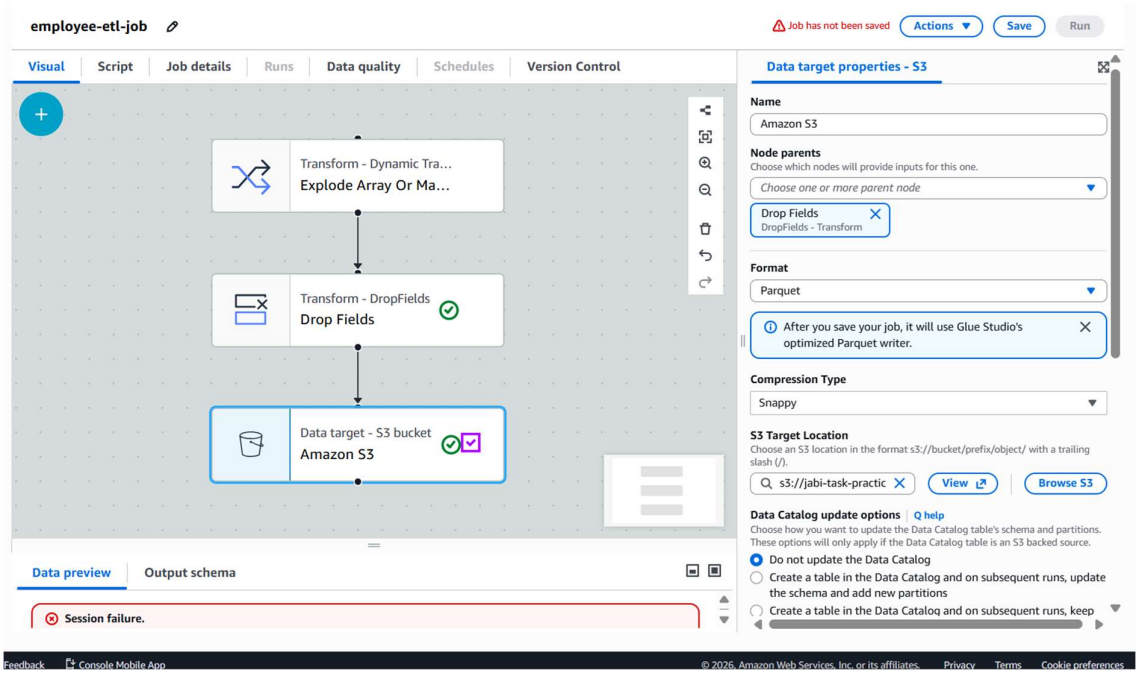
S3 Target Configuration

S3 target

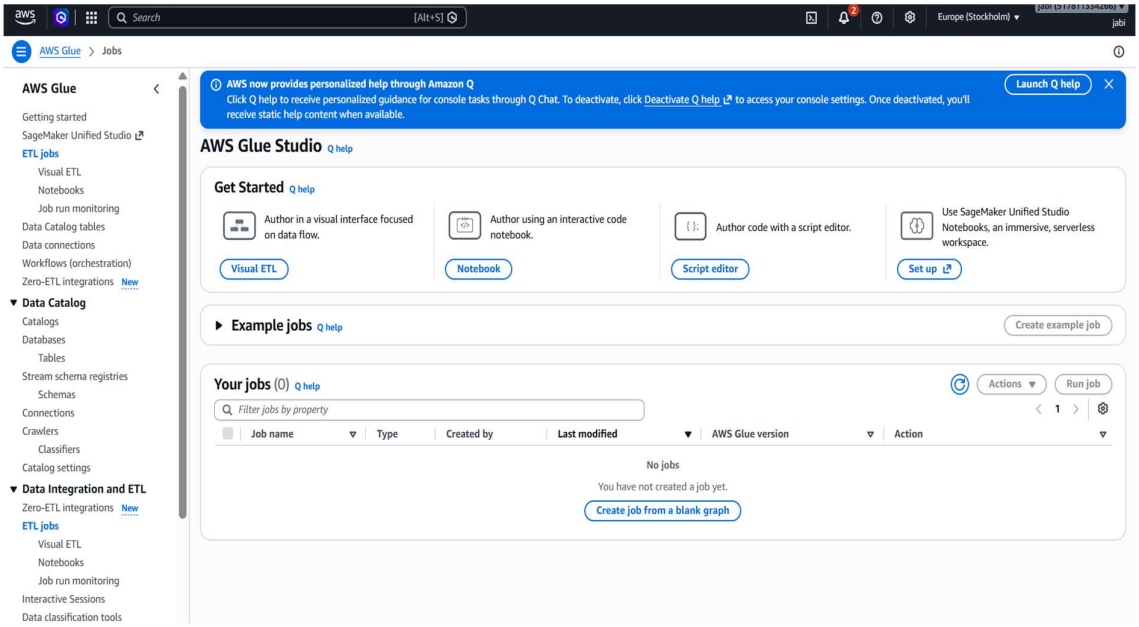




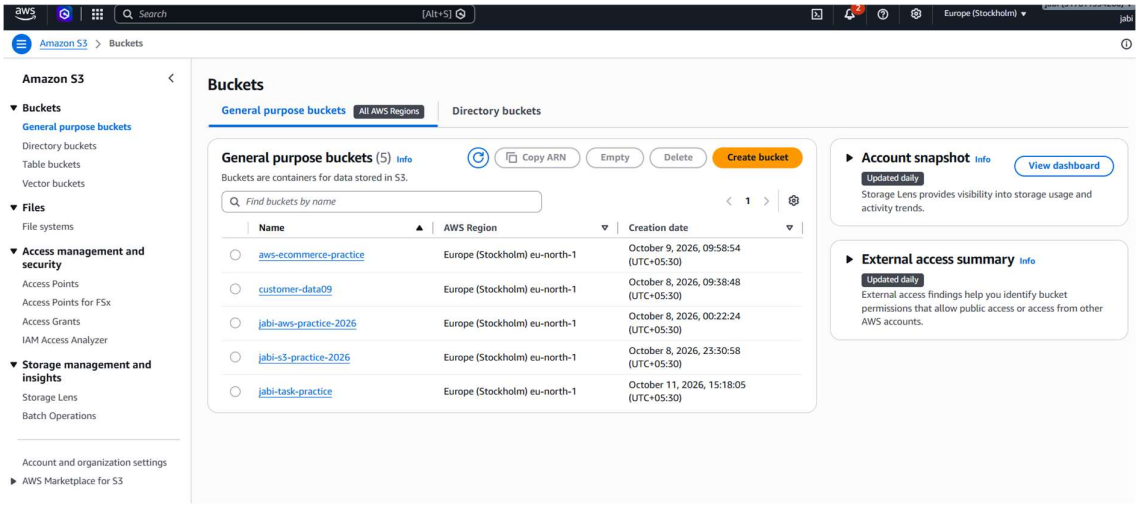
Relevant target configuration



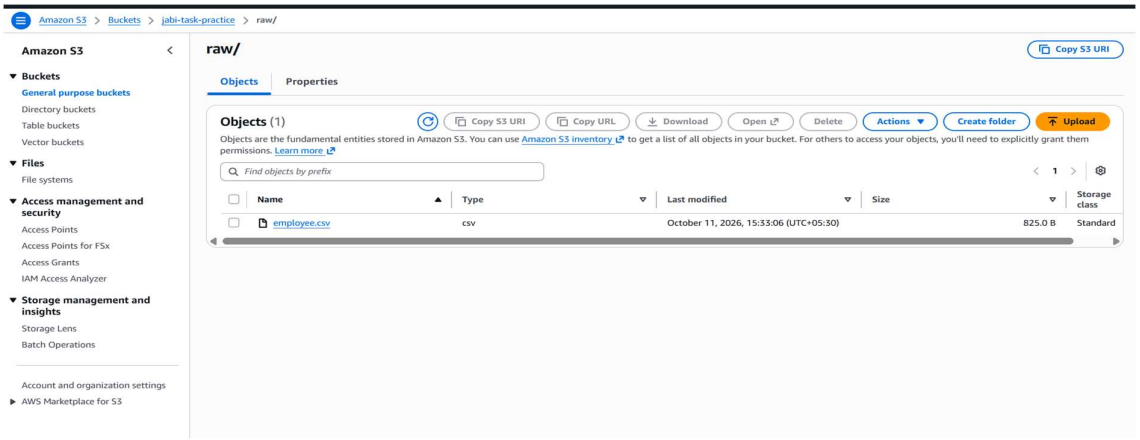
Glue ETL Job Execution



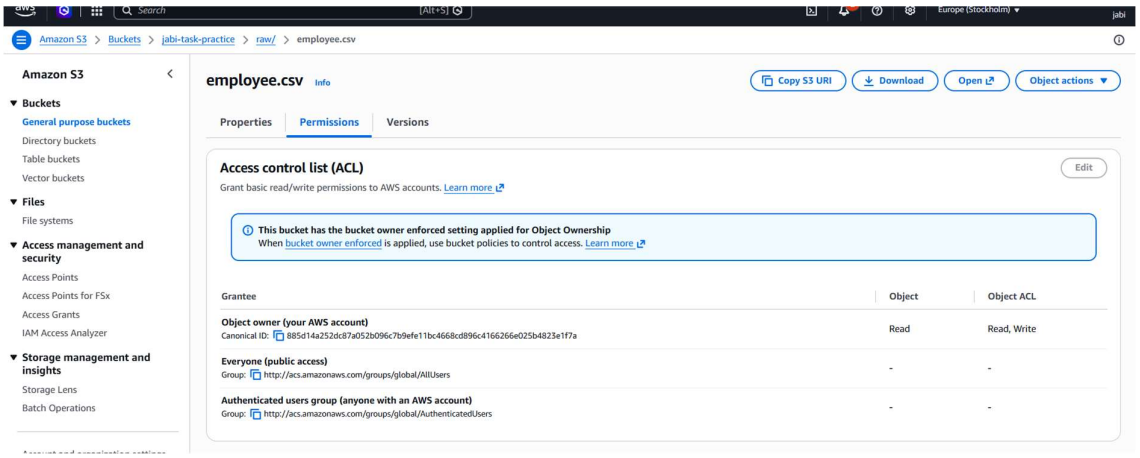
Final S3 Output



Generated output files



Parquet output



Final Explanation:

S3 : We upload the employee CSV file into the raw/ folder in the S3 bucket.

Glue Crawler: The crawler scans the CSV file and identifies its columns and data types.

Glue Data Catalog: The crawler stores the table details and schema in the Data Catalog.

Glue Visual ETL: We create a visual job to read the source data, transform it, and save the output.

Transformation: We use the Drop Fields transformation in AWS Glue to remove the city column from the employee data.

S3 Target: We configure Amazon S3 as the destination for the transformed data.

S3 Curated Data: The cleaned data is stored in the curated/ folder in S3.